

Day 8: Website Hosting via Docker on AWS EC2

Running Dockerized Web Servers on EC2 Infrastructure

Module 1: Docker Host Architecture

Running Docker containers inside virtual instances (like AWS EC2) combines the scalability of cloud resources with the portability of containerized workloads. It is the primary deployment method for microservices before moving to complex orchestrators like Kubernetes.

- **Isolation:** Processes run in isolated user spaces (namespaces) on the host, preventing version conflicts.
- **Portability:** The compiled image runs identically on local computers, test instances, and production servers.
- **Resource Efficiency:** Multiple containers run on a single EC2 instance without guest OS overhead.

Module 2: Practical Web Server Deployment Walkthrough

This walkthrough guides you through launching an EC2 instance, installing Docker, building a custom Nginx image, and hosting a live containerized website.

Step-by-Step Implementation Guide

- **1. Launch EC2 Instance:** Launch an EC2 instance with Ubuntu Server 22.04 LTS. Ensure Security Group allows HTTP (Port 80) and SSH (Port 22) traffic from anywhere.
- **2. Connect & Install Docker:** SSH into your EC2 instance using Git Bash and run the installation script:
`sudo apt update && sudo apt install docker.io -y`
- **3. Create Workspace Folder:** Create a new directory for your website resources:
`mkdir my-web-app && cd my-web-app`
- **4. Create Website Content:** Write a simple HTML index file:
`echo '<h1>Hosted via Docker on EC2!</h1>' > index.html`
- **5. Write the Dockerfile:** Create a Dockerfile in the same folder:
`FROM nginx:alpine
COPY index.html /usr/share/nginx/html/
EXPOSE 80`
- **6. Build Docker Image:** Compile the Docker container image:
`sudo docker build -t my-website-img .`
- **7. Run Web Container:** Launch the container in background daemon mode, mapping host port 80 to container port 80:
`sudo docker run -d -p 80:80 --name web-container my-website-img`
- **8. Verify Site Access:** Paste your EC2 instance's public IP address into a web browser. The page will load instantly.